Gnana Gopal Adusumilli

Software Engineer · Full-Stack · Cloud · Regulated (GxP) Systems
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SUMMARY

Software Engineer with 5+ years of experience in full-stack development, cloud architecture, and agile methodologies. Proven track record of improving application performance, implementing secure solutions, and delivering scalable applications using modern technologies. Expertise in microservices architecture, CI/CD automation, and cross-functional team collaboration.

SKILLS

Languages: Java, JavaScript, Python, SQL, PHP, HTML5, CSS3, Golang
Frameworks & Libraries: Spring Boot, React.js, Node.js, Express.js, Flask, Bootstrap, PySpark
Cloud Platforms: AWS, GCP, Azure
DevOps & Tools: Docker, Jenkins, Terraform, Git, Maven, Postman, Jira
Databases: MySQL, MongoDB, PostgreSQL, Elasticsearch, Teradata SQL
Business Intelligence: Power BI, Tableau, Birst, QuickSight, Splunk
Design & Documentation: Figma, ARIS, SAP SOLMAN

EXPERIENCE

Software Engineer — Oriontek Inc **May 2025 – Present**
Frederick, MD

- Led development of a Clinical Trial Inventory Management System (CTIMS) using React.js, Spring Boot, Flask, and Bootstrap - Built end-to-end receiving module handling 1000+ pharmaceutical shipments monthly with multi-package inspection workflows
- Architected auto-save system reducing data loss by 95% - Implemented debounced state management with React hooks and Python-driven validation pipelines for seamless user experience
- Designed role-based access control for 5 user roles using Spring Security with OAuth2 and JWT authentication - Enabled granular permissions across 15+ features ensuring compliance with pharmaceutical regulations
- Integrated AWS S3 with Docker-containerized microservices for document management - Built secure photo upload system handling discrepancy documentation, temperature logger graphs, and MySQL-backed compliance records
- Developed real-time inspection wizard with 7-section workflow using Golang microservices and Flask APIs - Streamlined receiving operations for warehouse staff, reducing inspection completion time by 25%
- Built admin dashboard with Splunk-based audit logging and Tableau, Birst, and AWS QuickSight reporting - Enabled tracking of 100,000+ user actions for FDA 21 CFR Part 11 compliance
- Implemented temperature excursion tracking system using Python, PySpark, and Teradata SQL - Automated cold chain compliance monitoring with alerts and segregation workflows for pharmaceutical materials

Software Engineer — HCL Global Systems Inc **Nov 2024 – Apr 2025**
Remote, OH

- Engineered high-performance application components using React.js, Spring Boot, Flask, and MySQL, reducing response times by 30% for a system serving 50K+ daily users.
- Architected and implemented microservices-based cloud infrastructure on AWS, utilizing ECS, EKS, Lambda, and Docker containerization to achieve 99.9% system uptime.
- Developed data processing pipelines using Python and PySpark to extract, transform, and load data from Teradata SQL databases, improving data retrieval efficiency for analytical reporting.
- Built interactive dashboards and business intelligence reports using Tableau, Birst, and AWS QuickSight, enabling stakeholders to make data-driven decisions with real-time insights.
- Configured Splunk for centralized log monitoring, application performance tracking, and real-time alerting, reducing mean time to incident resolution.
- Developed responsive front-end interfaces using Bootstrap for consistent cross-browser UI design and built high-performance backend services using Golang for concurrent processing tasks.
- Implemented automated CI/CD pipelines using Jenkins and AWS CodePipeline, reducing deployment time by 40%.
- Collaborated with cross-functional teams of 8+ members using agile methodologies and Git-based version control.

Information Technology Intern — Changing The Present **Feb 2024 – Nov 2024**

New York, NY

- Created and maintained a full-stack application using React.js, Bootstrap, Java Spring Boot, Flask, Node.js, HTML, and CSS, resulting in a 25% increase in user engagement and a 20% increase in income in six months.
- Developed data pipelines using Python and PySpark to process and transform datasets, and connected to Teradata SQL databases for efficient data storage and retrieval operations.
- Built reporting dashboards using Tableau, Birst, and AWS QuickSight to visualize donation trends, user engagement metrics, and platform performance for stakeholders.
- Developed custom Docker images and pushed them to DockerHub and Amazon ECR, enabling reproducible builds and versioned microservice rollouts.
- Configured Splunk for application log aggregation, error monitoring, and performance analysis, improving issue detection and platform reliability.
- Secured APIs and user sessions by combining Spring Security and Node.js (Express.js) with OAuth2 and JWT authentication to improve application security and access management.
- Developed backend microservices using Golang for high-throughput concurrent processing and lightweight API endpoints.
- Implemented Redis caching and conducted CRUD operations with relational databases (MySQL, PostgreSQL) using Spring Data JPA and Mongoose for MongoDB.

Information Technology Intern — Changing The Present **Aug 2023 – Nov 2023**

New York, NY

- Revolutionized charity gift purchase platform using React.js, Redux, and Bootstrap, implementing OAuth 2.0 authentication, resulting in 56% increase in user subscriptions.
- Developed ETL pipeline using Java Spring Boot and Python to integrate Shopify data with Power BI and Tableau, improving data processing efficiency by 65%.
- Orchestrated containerized application deployment using Docker and Azure App Service, achieving zero-downtime deployments.
- Designed and implemented RESTful APIs using Node.js, Express.js, and Flask, integrated with MongoDB for scalable data management.
- Automated deployment workflows using GitHub Actions, reducing release cycle time by 40% and improving code quality.

Application Development Analyst — Accenture **Sep 2019 – Jul 2022**

Bangalore, India

- Spearheaded migration of legacy application to Spring Boot microservices architecture, improving performance by 17% for 25,000 monthly users.
- Gathered and documented business requirements using JIRA and Microsoft Visio, coordinating with stakeholders across APAC, EMEA, LATAM, and NA regions.
- Developed automated reporting system using JavaScript and Python, reducing manual analysis time by 70%.
- Implemented automated testing framework reducing deployment time by 45%, achieving 98% test coverage.
- Utilized AWS CloudWatch and Trusted Advisor to optimize cloud resource utilization, reducing infrastructure costs by 25%.

PROJECTS

- **Data Analysis on COVID-19 Dataset:** Analysis of pandemic trends and impacts, published in the International Journal of Advanced Research in Computer and Communication Engineering.
- **Exploratory Data Analysis on Crime Data:** Analysis of 84.51° data to derive insights on specific crime types using Databricks, Spark, Python, and SQL.
- **Credit Card Fraud Detection:** Fraud detection system using Python with Logistic Regression and Decision Tree classifiers.
- **House Price Prediction System:** End-to-end ML project with EDA, feature engineering, testing, and ZenML + MLflow for experiment tracking and deployment.

EDUCATION

University of Cincinnati — Master of Science in Information Technology (Aug 2022 – Dec 2023) · GPA: 4.0/4.0
Visvesvaraya Technological University — Bachelor of Engineering in Computer Science (Aug 2015 – Aug 2019)

CERTIFICATIONS

AWS Certified Developer – Associate • AWS Certified Cloud Practitioner • Microsoft Azure Administrator Associate (AZ-104) • Microsoft Azure Fundamentals (AZ-900)